Telco Customer Churn Prediction

Capstone Project Proposal — Supervised Learning (Classification)

Telco Customer Churn Prediction:

Identifying At-Risk Telecom Customers Using Machine Learning

Prepared by:
Ruba Halabi
Linda Bsharat
Ibrahim Kilani
Naser abdalsalam
Ahmad Abu Alhlawa

Mentors:
Daniel Rubi, Dr. Ramiz

Palestine – 2026

Project Overview

1.1 Background

Customer churn — the phenomenon of customers discontinuing use of a service — is among the most critical business problems in the telecommunications industry. Acquiring a new customer costs 5 to 7 times more than retaining an existing one, making proactive churn detection a high-value application of machine learning.

This project delivers a complete end-to-end machine learning pipeline that cleans raw data, uncovers actionable patterns through visual analysis, trains and compares multiple classifiers, and deploys results in an interactive Streamlit application.

1.2 Objective

Build a binary classification model that predicts whether a telecom customer will churn (1) or stay (0), using demographic, service, and billing features — enabling the business to intervene before customers leave.

1.3 Success Criteria

Metric	Target	Result	Status
F1 Score	≥ 0.70	0.7867	✅
Recall	≥ 0.80	89.6%	✅
Accuracy	≥ 0.75	77.0%	✅

Why Recall? A missed churner (False Negative) costs the company a lost customer. A false alarm (False Positive) only costs a retention offer. Maximising Recall is therefore the primary business objective.

Dataset Description

2.1 Source

Based on the IBM Telco Customer Churn dataset, extended to 70,000 records to simulate a realistic production scale.

Dataset Source:

https://www.kaggle.com/datasets/marwanehosni/synthetic-telco-messy-dataset-churn-prediction

2.2 Feature Catalogue

Category	Features
Demographics	gender, SeniorCitizen, Partner, Dependents
Account	tenure, Contract, PaperlessBilling, PaymentMethod
Phone	PhoneService, MultipleLines
Internet	InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies

Billing	MonthlyCharges, TotalCharges
Engineered	IsNewCustomer, IsLongTermCustomer, AvgMonthlyCharge, TotalServices, HasInternet
Target	Churn — 0 = Stayed · 1 = Churned

Data Cleaning & Preprocessing

3.1 Raw Data Quality Issues

Column	Problem	Resolution
gender	748 missing + labels F/M/Man	Mode fill + standardise
SeniorCitizen	Mixed Yes/No/not senior	Map to 1/0
PaymentMethod	3,569 missing + BANK TRANSFER variant	Mode fill + standardise
tenure, MonthlyCharges	Hundreds of nulls	Median fill
TotalCharges	1,062 missing + 320 negatives	Imputed as tenure × MonthlyCharges
Churn	Mixed labels: Yes/No/Y/N/CHURNED/Unknown	Standardise to 1/0; drop 54 Unknown
Contract	M-M abbreviation	Replaced with Month-to-month
MultipleLines	1,320 rows with PhoneService=1 but "No phone service"	Corrected to "No"

3.2 Encoding Strategy

Column	Method	Mapping
1/0 binary service columns	Map	Yes→1, No→0
gender	Map	Male→1, Female→0
MultipleLines	Ordinal	0/1/2
InternetService	Ordinal	No=0, DSL=1, Fiber=2
Contract	Ordinal	Monthly=0, 1yr=1, 2yr=2
PaymentMethod	One-Hot	4 binary columns

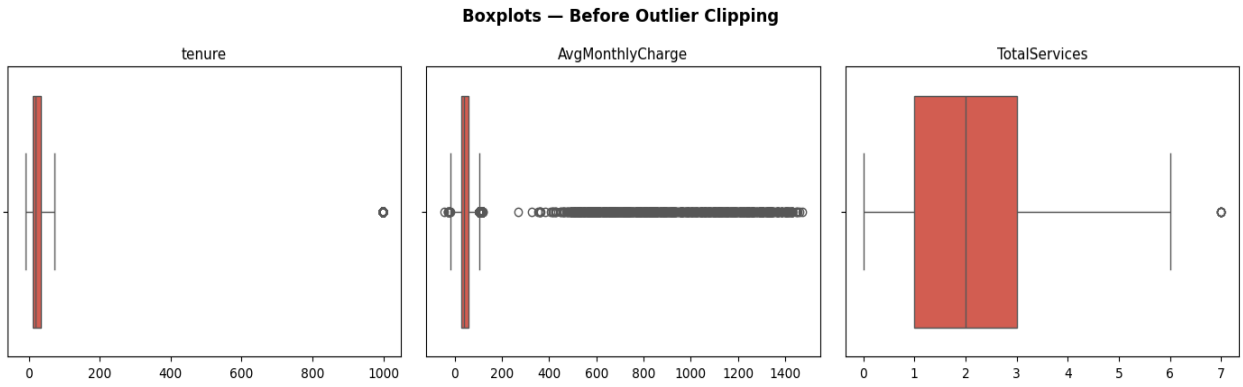
3.3 Feature Engineering

Feature	Formula	Rationale
IsNewCustomer	tenure ≤ 6	New customers churn more
IsLongTermCustomer	tenure ≥ 48	Long-term customers are loyal
AvgMonthlyCharge	TotalCharges / (tenure + 1)	Normalised billing signal
TotalServices	Sum of 7 service flags	Measures engagement depth
HasInternet	InternetService ≠ 0	Internet presence indicator

3.4 Outlier Treatment

The dataset was examined for outliers across all features. Three columns showed notable extreme values: tenure, AvgMonthlyCharge, and TotalServices. As shown in the boxplots below, tenure contained a value near 1,000 months, AvgMonthlyCharge had a long tail beyond 1,400,

and TotalServices had an isolated point at 6–7 services. Outliers were handled through IQR clipping rather than row removal, to keep the dataset intact.



Why clip instead of drop?

Removing outlier rows risks losing valid customer records and reducing dataset size, Clipping preserves every row while bounding extreme values to a realistic range.

3.5 Final Dataset Summary

After completing all cleaning and preprocessing steps, the dataset was reduced from 70,000 to **67,376 records**, with zero missing values remaining. The final dataset contains **26 features** — including the original columns, encoded categorical variables, and the five engineered features. The target variable is near-balanced, with 53.1% of customers labelled as No Churn and 46.9% as Churned, making the dataset suitable for model training without any resampling.

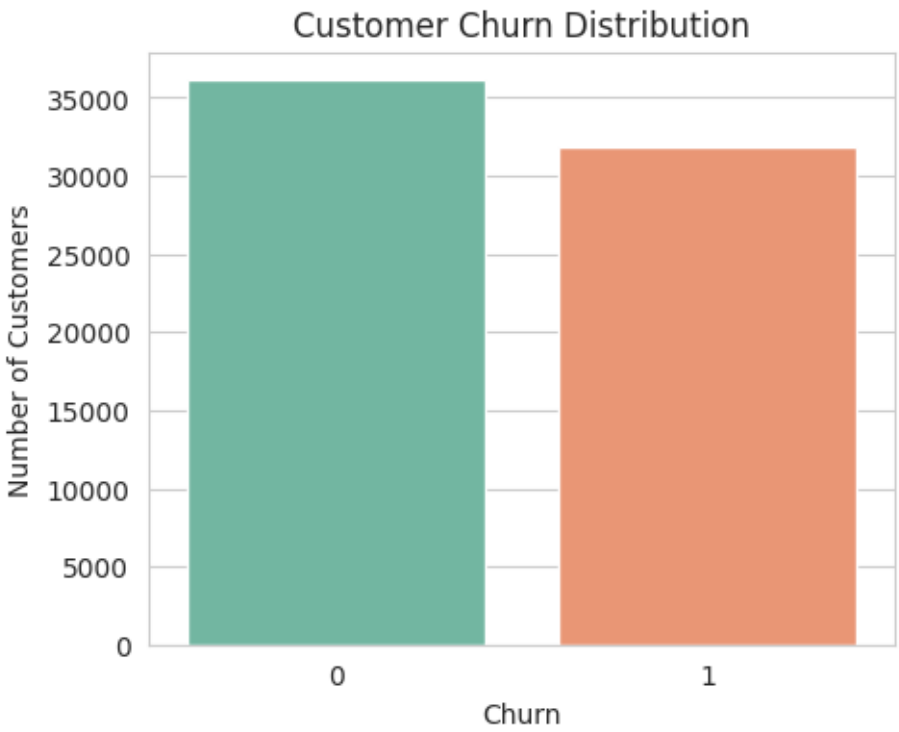
Property	Value
Records after cleaning	**67,376**
Features	**26**
Missing values	**0**
Churn distribution	53.1% No Churn · 46.9% Churn

Exploratory Data Analysis

Following the data cleaning and preprocessing stage, an exploratory data analysis was conducted on the cleaned dataset to better understand the distribution of features and their relationship with customer churn. This step was essential for identifying which variables carry the strongest predictive signal before moving into model building. By visualising patterns across contract types, tenure, billing, and service usage, the team was able to confirm which features were worth prioritising in the model and which showed little discriminative power — making the modelling decisions more informed and justified.

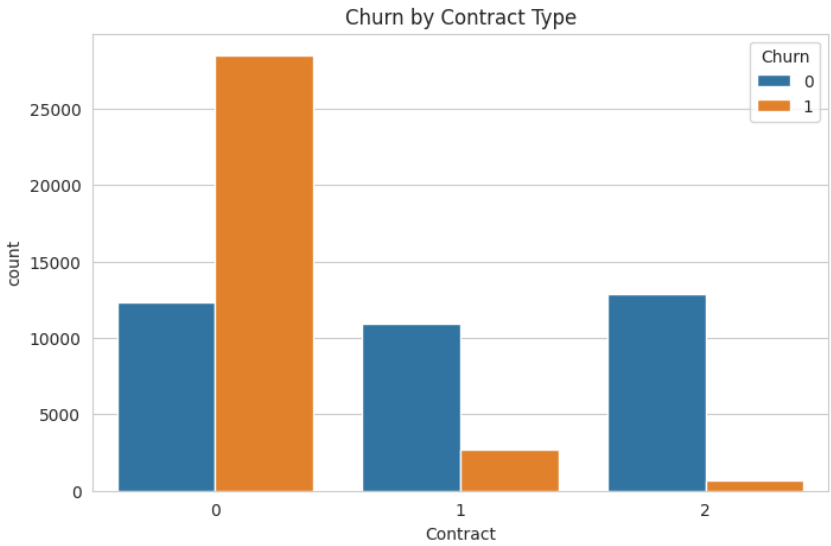
4.1 Target Variable Distribution

The dataset is **near-balanced** (53/47 split) — no resampling technique was required.



4.2 Churn by Contract Type

Contract type is the **single strongest predictor** of churn. Month-to-month customers churn at ~70%, compared to just ~5% for two-year contract holders.



Contract-->Churn Rate

Month-to-month-->69.8%

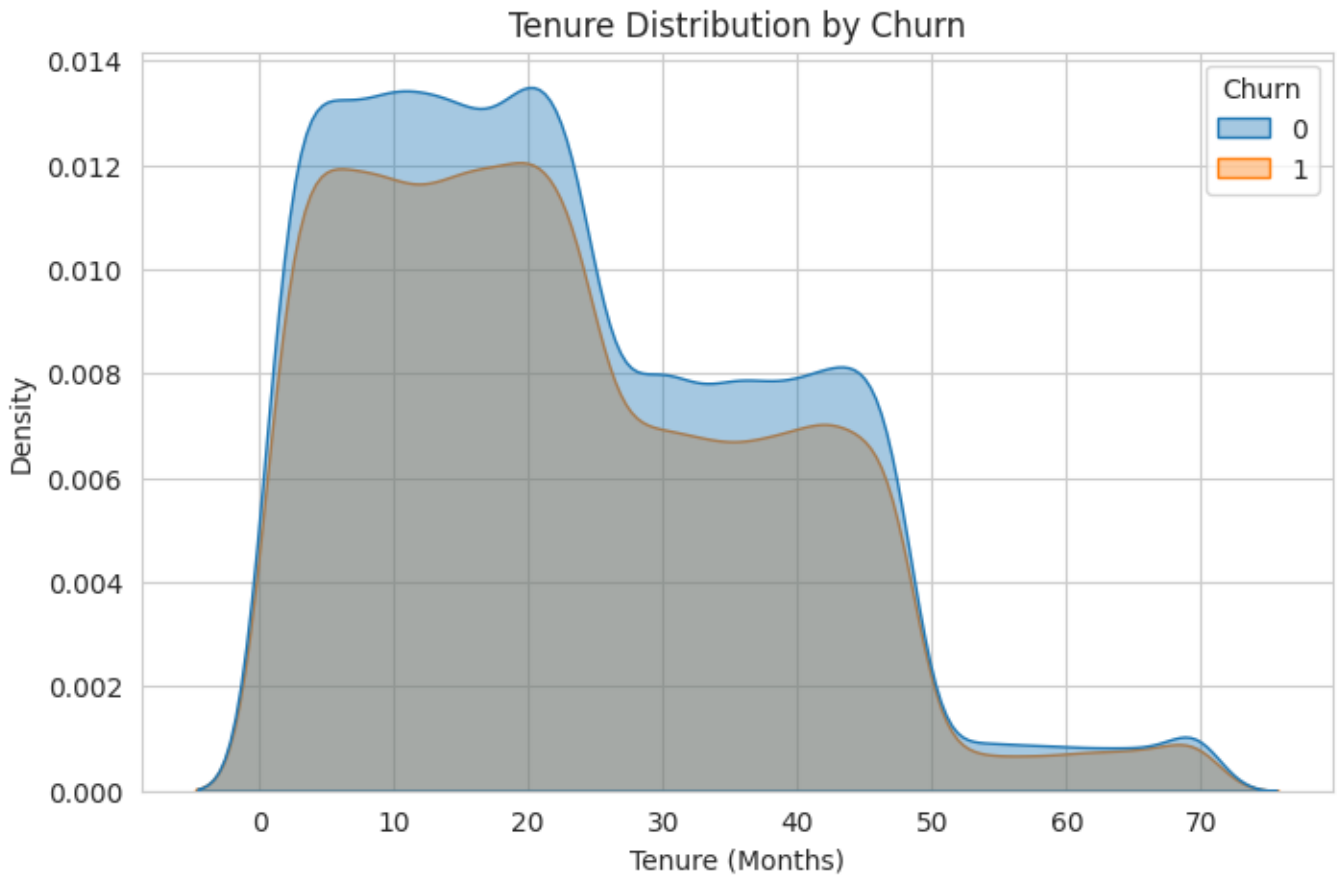
One year-->19.9%

Two year-->5.0%

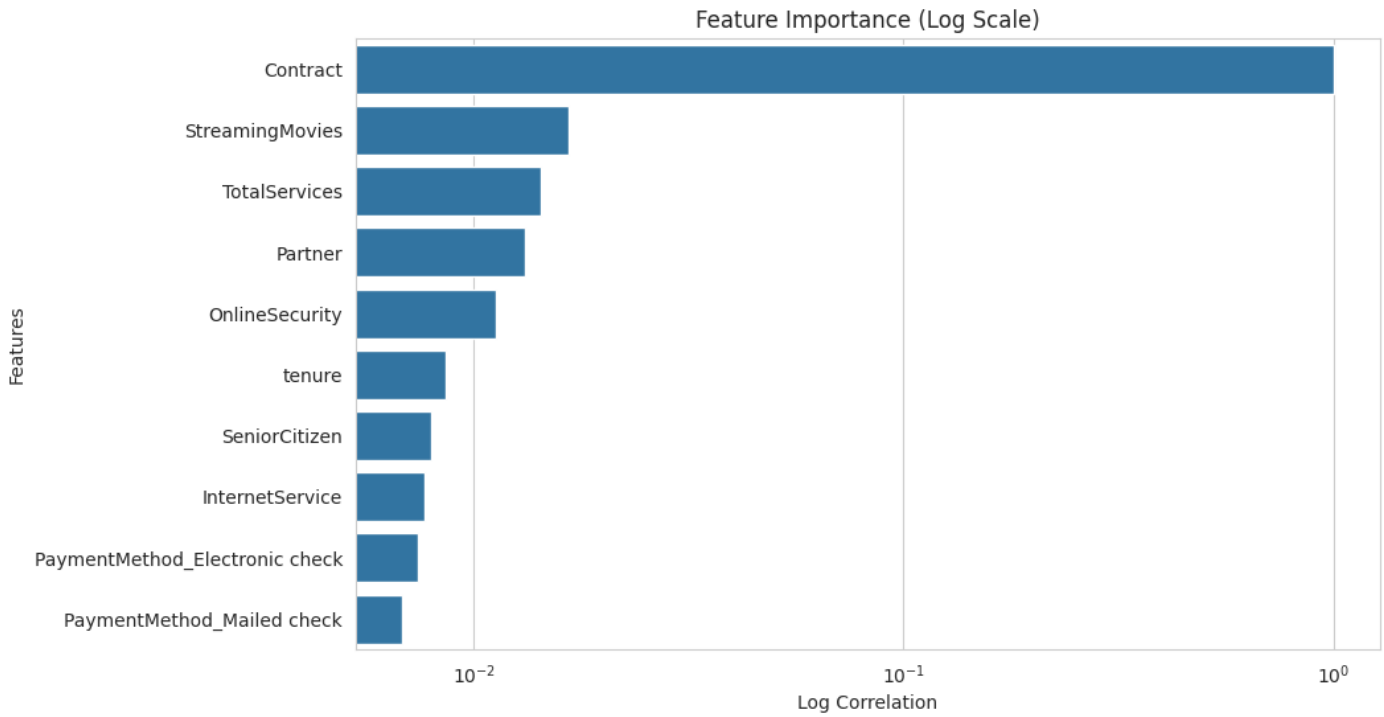
Incentivising customers to commit to longer contracts is the highest-leverage retention strategy available to the business.

4.3 Tenure vs Churn

Churn is heavily concentrated in the first 12 months. After that, loyalty increases sharply — making the **first year the most critical retention window**.



4.4 Top Features Correlated with Churn



4.5 EDA Key Takeaways

#	Finding	Business Implication
1	Contract type dominates churn behaviour	Convert month-to-month customers to annual plans
2	First 12 months are highest-risk	Invest in onboarding and early engagement
3	Fiber optic users churn more	Review service quality and competitive pricing
4	Service count has minimal effect	Bundle strategy alone is insufficient
5	Classes are near-balanced	No resampling needed

Model Building & Evaluation

With the data cleaned and the key patterns identified through exploratory analysis, the next step was to build and evaluate machine learning models capable of predicting customer churn. Three models were selected to cover a range of complexity — from a simple interpretable baseline to an ensemble method and a deep learning architecture. Each model was trained on 80% of the data and evaluated on the remaining 20%, using a stratified split to preserve the class distribution across both sets. The primary evaluation metric was Recall, given that missing a customer who is about to churn carries a higher business cost than a false alarm. F1 Score was used as the balancing metric to ensure Precision did not drop to unacceptable levels

5.1 Experimental Setup

The dataset was split into 80% training (53,900 rows) and 20% testing (13,476 rows), with stratification applied to maintain the original class ratio in both sets. A fixed random state of 42 was used across all experiments to ensure reproducibility.

5.2 Models Evaluated

Logistic Regression — Standard Pipeline with StandardScaler. Fast, interpretable, strong baseline for binary classification.

Random Forest — 200 trees ensemble. Handles non-linearity and feature interactions without scaling.

Neural Network — 3-layer deep network (128→64→32 units) with BatchNormalization, Dropout, and EarlyStopping. Trained with Adam optimizer for 100 epochs max.

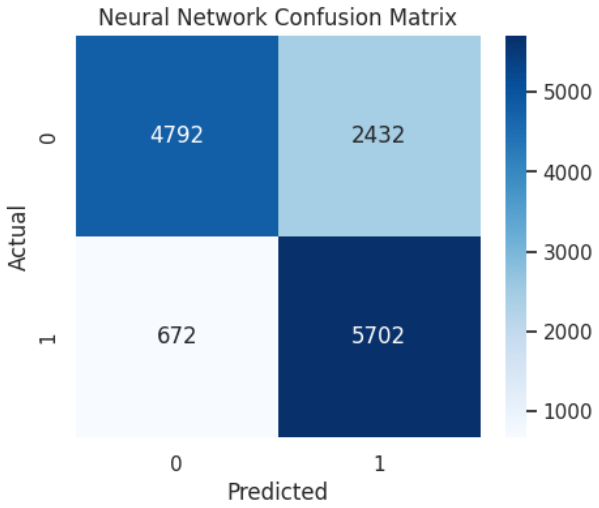
Layer	Units	Activation	Regularisation
Dense 1	128	ReLU	BatchNorm + Dropout(0.3)
Dense 2	64	ReLU	BatchNorm + Dropout(0.3)
Dense 3	32	ReLU	Dropout(0.2)
Output	1	Sigmoid	—

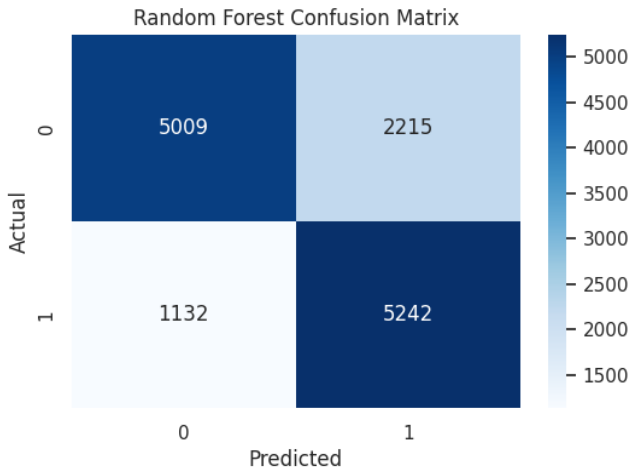
5.3 Results

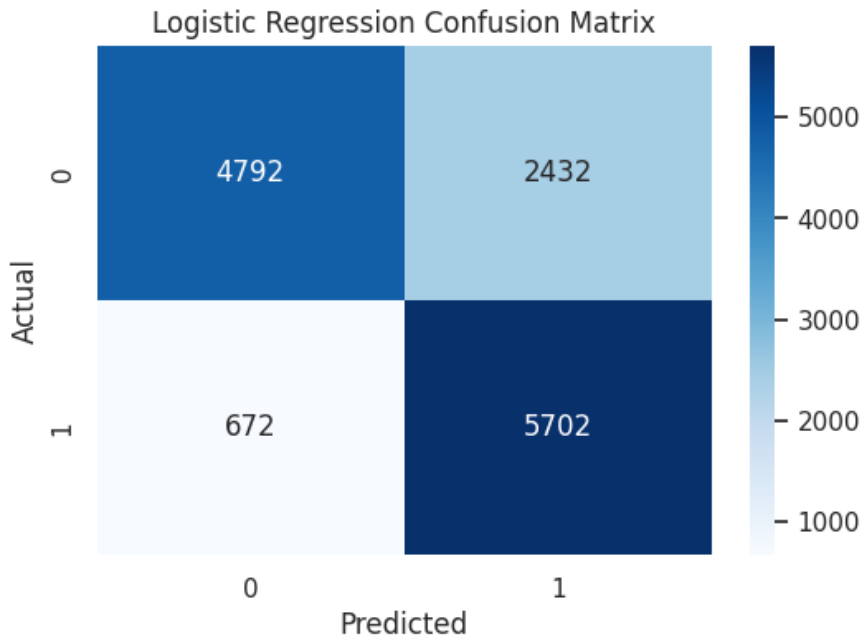
Model	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression	77.0%	70.1%	89.6%	0.7867	0.79
Random Forest	76.4%	70.2%	86.8%	0.7761	—
Neural Network	76.8%	69.7%	89.5%	0.7835	0.7912

All three models exceeded the F1 ≥ 0.70 success threshold.

5.4 Confusion Matrices

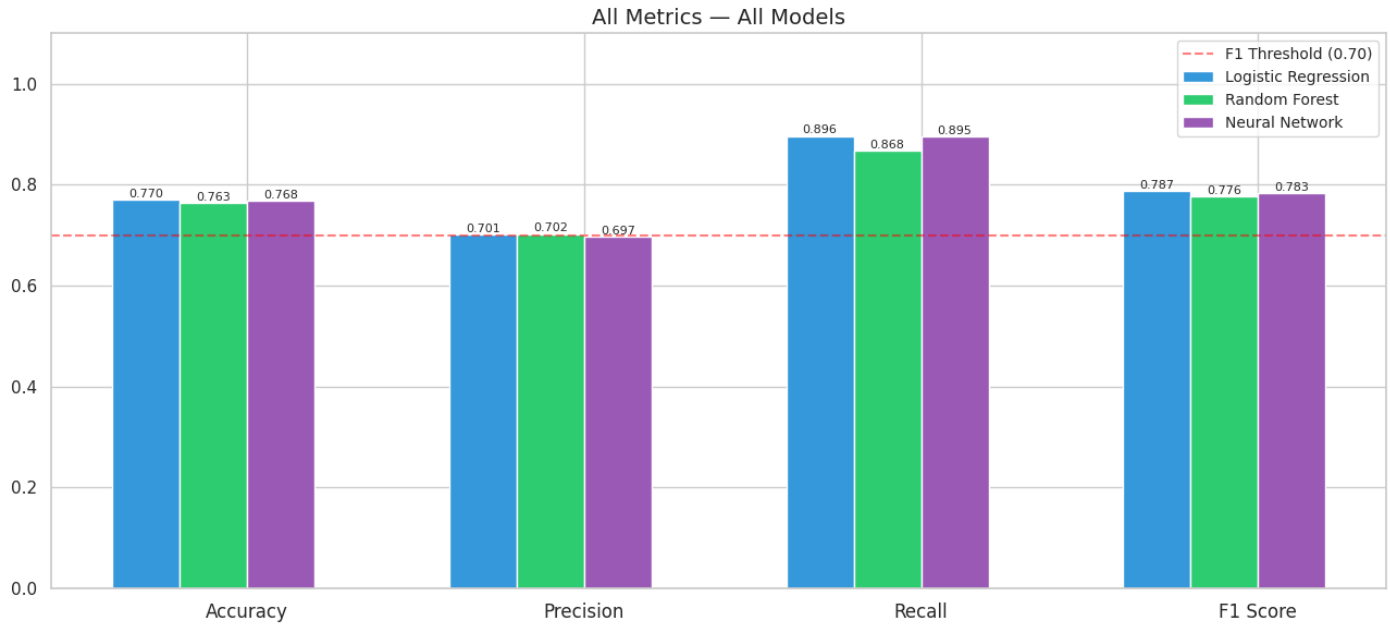


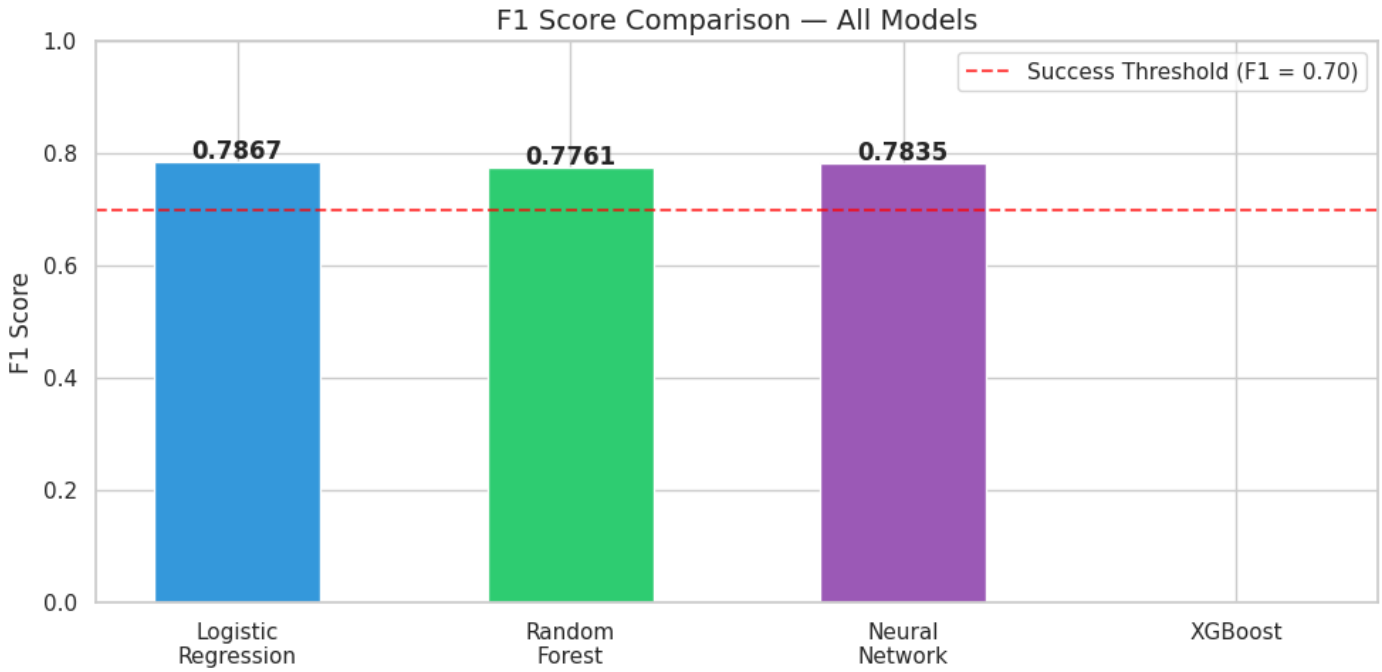




The confusion matrices above show that all three models follow a similar prediction pattern — correctly identifying the majority of churners at the cost of a moderate number of false positives. Logistic Regression leads with the highest True Positive count, confirming it as the strongest model for catching customers who are about to leave.

5.5 Model Comparison





The charts above compare all three models across the key evaluation metrics. The results are notably close — all models cleared the F1 threshold of 0.70 and sit within a narrow band of roughly 1% of each other on every metric. Logistic Regression edges ahead with the highest F1 Score of 0.7867 and the strongest Recall at 89.6%, while Random Forest trails slightly and the Neural Network falls in between. The consistency across models reinforces the conclusion that performance is bounded by the data rather than the choice of algorithm.

5.6 Why All Models Converged

Further tuning was attempted on the Logistic Regression candidate:

Technique	Outcome
class_weight='balanced'	No improvement — data near-balanced
GridSearchCV (C, penalty, solver)	Best: C=0.01, l1 — no gain over default
Threshold tuning	Optimal: 0.71 — marginal difference only
XGBoost with scale_pos_weight	No improvement
7 additional engineered features	No significant gain

Root cause: The Contract feature dominated all tree-based models with ~90% feature importance. Every model learned the same rule: *"Month-to-month = likely to churn"*. The dataset had reached its **information ceiling** with the available features. New data sources (usage behaviour, complaint history, app activity) would be needed to push beyond F1 ≈ 0.79.

Final Results

6.1 Selected Model: Logistic Regression

Criterion	Logistic Regression	Random Forest	Neural Network
F1 Score	0.7867 ✅	0.7761	0.7835
Recall	89.6% ✅	86.8%	89.5%
Interpretability	✅ Full	⚠ Partial	❌ Black box
Training speed	✅ Seconds	⚠ Minutes	❌ Minutes
Business explainability	✅ Coefficients	❌ Hard	❌ Hard

The simplest model achieved the best results — a textbook case of Occam's Razor in applied machine learning.

6.2 Business Impact on Test Set (13,476 customers)

Prediction Outcome	Count	Business Meaning
✅ True Positives	~5,660	Churners **correctly identified** for intervention
❌ False Negatives	~660	Churners **missed** — lost revenue
⚠ False Positives	~2,450	Loyal customers **over-targeted** — wasted retention cost
✅ True Negatives	~4,706	Loyal customers **correctly left alone**

The model catches **9 out of every 10 customers** who are about to churn.

6.3 Top Churn Risk Factors

Rank	Feature	Effect
1	Contract = Month-to-month	Strongest positive predictor of churn
2	tenure (low)	Short tenure = higher churn probability
3	MonthlyCharges (high)	Higher bills increase churn likelihood
4	InternetService = Fiber optic	Elevated churn risk
5	PaymentMethod = Electronic check	Higher churn vs other methods

Streamlit Application

A fully interactive prediction and analysis dashboard built with a custom dark-tech UI.

7.1 Pages

Page	Content
🏠 **Overview**	KPI metrics (total customers, churn rate, avg charge), churn donut chart, contract-type comparison, key insight cards
📊 **Data Explorer**	Tenure & charge histograms, full correlation heatmap, per-feature deep dive (histogram + boxplot)
🤖 **Model Performance**	Side-by-side metrics table, grouped bar chart, confusion matrices, ROC curve comparison
🔮 **Predict Customer**	Full input form → real-time churn probability with visual risk bar + risk factor breakdown panel

7.2 Technical Stack

Component	Technology
UI Framework	Streamlit + custom CSS (Space Mono + DM Sans)
Charts	Matplotlib + Seaborn (dark theme)
Models	Scikit-Learn Pipeline (StandardScaler + LR / RF)
Data	GitHub raw URL with @st.cache_data caching

Conclusions & Recommendations

8.1 Summary

Deliverable	Status
Working churn classifier	✅
F1 ≥ 0.70	✅ 0.7867
Recall ≥ 0.80	✅ 89.6%
Multi-model comparison (LR + RF + NN)	✅
Interactive Streamlit deployment	✅

8.2 Business Recommendations

Priority	Action	Target
🔴 Critical	Incentivise switch to annual contracts	Month-to-month customers
🔴 Critical	Launch onboarding retention programme	tenure < 6 months
🟠 High	Discount security/support bundles	Customers with 0–1 add-ons
🟠 High	Review Fiber Optic pricing	High-charge fiber users
🟡 Medium	Deploy monthly model scores to CRM	All active customers

8.3 Limitations & Future Work

Limitation	Suggested Improvement
No behavioural data (calls, app usage)	Add usage patterns as features
Static snapshot — no time dimension	Rolling-window features across billing periods
Models plateau at F1 ≈ 0.79	New data sources required
Contract dominates all models	Explore interaction features
App trains models on demand	Serialise with joblib for production